KAIZEN MATHS

Lesson starter

HCF, LCM and Prime Factorisation

Basic Numbers

Name _____

S1

HCF(18, 22)

S2

LCM(10, 12)

S3

HCF(30, 27)

S4

LCM(18, 22)

Show your working

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NUM-A60B-L1 · S

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KAIZEN MATHS

Exit ticket

HCF, LCM and Prime Factorisation

Basic Numbers

Name _____

E1

HCF(34, 20)

E2

LCM(21, 28)

E3

HCF(28, 14)

KAIZEN MATHS

Exit ticket

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Teacher answers & working

HCF, LCM and Prime Factorisation · Basic Numbers

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S1

HCF(18, 22)

Answer:

2

Working:

Find the HCF of 18 and 22

Prime factorisation of 18: 2×3^2

Prime factorisation of 22: 2×11

Common prime factors: 2

Multiply the common prime factors: 2 = 2

HCF = 2

S2

LCM(10, 12)

Answer:

60

Working:

Find the LCM of 10 and 12

Prime factorisation of 10: 2×5

Prime factorisation of 12: $2^2 \times 3$

Use the highest power of every prime factor needed

Prime factors used for the LCM: $2^2 \times 3 \times 5$

Multiply these factors: $2^2 \times 3 \times 5 = 60$

LCM = 60

S3

HCF(30, 27)

Answer:

3

Working:

Find the HCF of 30 and 27

Prime factorisation of 30: $2 \times 3 \times 5$

Prime factorisation of 27: 3^3

Common prime factors: 3

Multiply the common prime factors: 3 = 3

HCF = 3

Teacher answers & working

HCF, LCM and Prime Factorisation · Basic Numbers

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S4

LCM(18, 22)

Answer:

198

Working:

Find the LCM of 18 and 22

Prime factorisation of 18: 2×3^2

Prime factorisation of 22: 2×11

Use the highest power of every prime factor needed

Prime factors used for the LCM: $2 \times 3^2 \times 11$

Multiply these factors: $2 \times 3^2 \times 11 = 198$

LCM = 198

E1

HCF(34, 20)

Answer:

2

Working:

Find the HCF of 34 and 20

Prime factorisation of 34: 2×17

Prime factorisation of 20: $2^2 \times 5$

Common prime factors: 2

Multiply the common prime factors: $2 = 2$

HCF = 2

Teacher answers & working

HCF, LCM and Prime Factorisation · Basic Numbers

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E2

LCM(21, 28)

Answer:

84

Working:

Find the LCM of 21 and 28

Prime factorisation of 21: 3×7

Prime factorisation of 28: $2^2 \times 7$

Use the highest power of every prime factor needed

Prime factors used for the LCM: $2^2 \times 3 \times 7$

Multiply these factors: $2^2 \times 3 \times 7 = 84$

LCM = 84

E3

HCF(28, 14)

Answer:

14

Working:

Find the HCF of 28 and 14

Prime factorisation of 28: $2^2 \times 7$

Prime factorisation of 14: 2×7

Common prime factors: 2×7

Multiply the common prime factors: $2 \times 7 = 14$

HCF = 14